CHEM 427/527 · SESSION 13 · THURSDAY, NOVEMBER 12, 2026

Curve fitting: does our size method pass the NIST standard?

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Drafted with assistance by Claude Code. Reviewed and revised by Matt.

BY THE END OF THIS SESSION YOU CAN

Five things to do with a fitted line

1. Fit a model function to measured points from a starting guess and bounds taken from the physics, and carry each parameter’s standard error to the quantity the decision needs.
2. Read a residual plot to tell a profile the data support from one they do not, and state what a laboratory that skipped that check would have reported.
3. Choose a peak profile from the mechanism that broadens the line, and show how far a profile chosen for convenience moves the result.
4. Decide whether a method passes validation against a certified value and its uncertainty, with the margin to the interval’s edge and whether a second profile changes the call.
5. Write a prompt that names the model, the weights, the bounds and the quantity to report, and check the answer against a cell that recomputes those quantities.

LAPTOPS, TO MINUTE 2

Run git pull; session 13 appears

In the terminal

```
git pull
```

The pull is larger than usual: the five patterns are 5.5 MB each.

What appears

The folder S13-curve-fitting, with data/ holding five .cif files, certificate.csv, windows.csv, reflections.csv and README.md. Nothing is downloaded today.

Tuesday’s session 12 notebook and today’s are both due at 1:30 pm on Tuesday, November 17.

THE DECISION

Does our size method pass the NIST standard?

Who decides

A laboratory validates its line-broadening method on NIST SRM 1979 before it reports nanoparticle sizes. It fits a profile to one diffraction line, takes the crystallite size from the fitted integral breadth by the Scherrer equation, and asks whether that size lands inside the certified interval for the reflection.

Passed wrongly

Every catalyst or pigment size the laboratory reports is biased by the same amount.

Failed wrongly

A method that works is sent back for revalidation.

Real data: the five patterns NIST measured to certify the 15 nm powder

Each pattern holds 45,504 points from 2.0 to 47.5° at a wavelength of 0.414 Å, measured on a synchrotron beamline. The certificate gives the size and its uncertainty for 22 reflections. You have one reflection of one pattern.

FROM A VERTICAL LINE TO A FITTED SHAPE

What does fitting two peak shapes give that the vertical line cannot?

Session 12

A doublet was split by a vertical line at its valley, and each half’s area counted for its peak.

Today

A shape is fitted to each peak: a function with a centroid, a width and an area.

The question

Two peaks overlap. What does the pair of fitted shapes tell you that the line at the valley does not?

Which number does the vertical line not have?

THE ANSWER

A width for each peak, and a tail assigned to its owner

The tail under the neighbour

A fitted shape runs under the other peak, so the area hidden under the neighbour is counted for the peak it belongs to.

The width

Each shape has a width of its own. Today’s crystallite size comes from that width, through the integral breadth and the Scherrer equation.

The vertical line

A line at the valley has no width and no tail. It gives an area, and nothing about the shape that made it.

The size you report today is the width of a fitted shape. The shape is the choice.

LIVE: CURVE_FIT ON DATASET A, THE 032 LINE OF PATTERN 5857

What did the fit with no starting guess fit?

No starting guess: every parameter starts at 1

19.46°

The fitted centroid, 7.68° from the line, with a width of 1.00° and an area of 1. The optimizer stopped where the gradient vanished, and returned numbers with a warning.

The guess from the physics

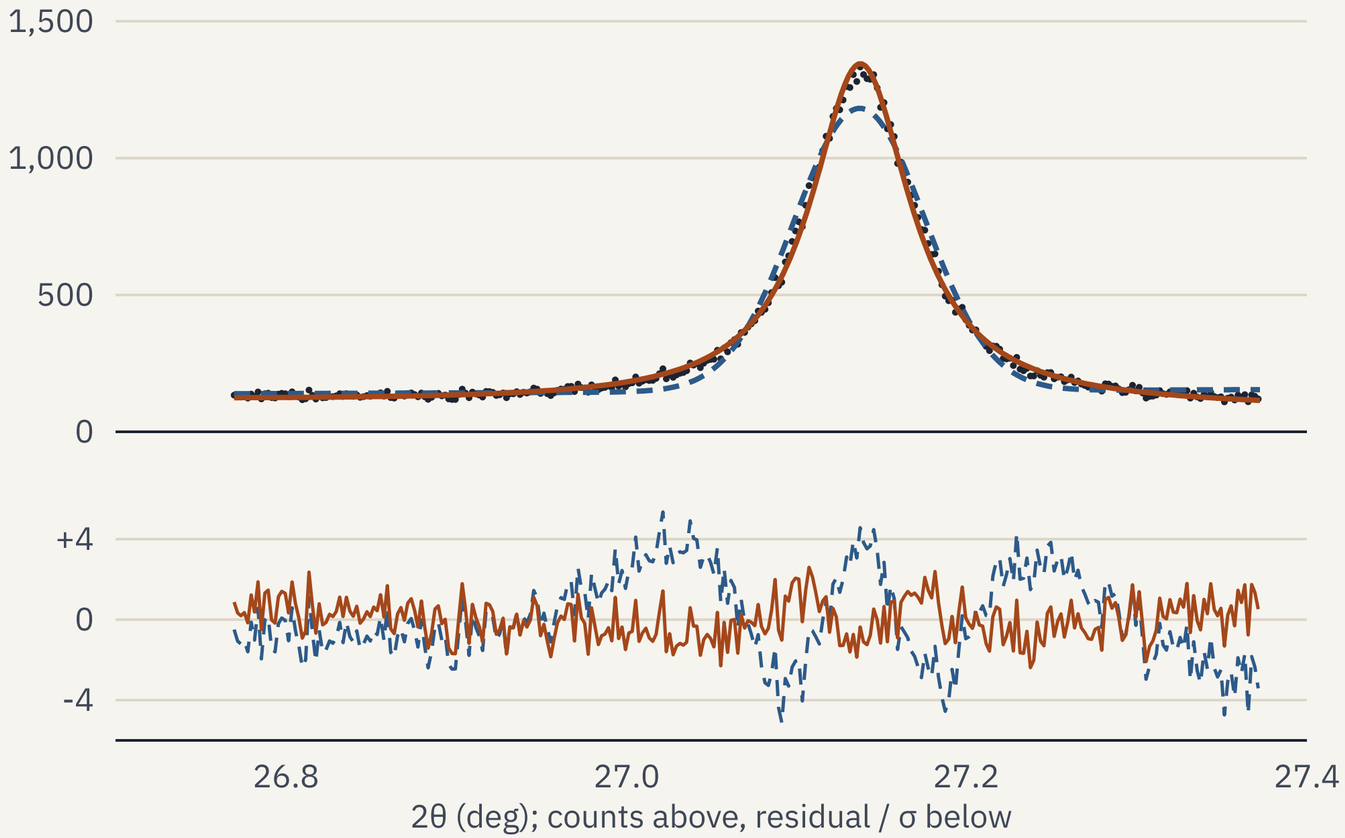
27.1417°

The centroid where the certified lattice puts the 032 reflection; the width 0.163°, which a 15 nm crystallite gives there; the area as height times width; the background near 129 from the window’s ends.

A fit finds the nearest minimum, not the best one. The guess says where to start, and the bounds say where it may go.

LIVE: TWO PROFILES ON DATASET A, FROM THE SAME STARTING GUESS

Which size does the certificate agree with?



Gaussian, dashed: L = 25.38 ± 0.09 nm, fail; reduced chi-squared 4.43

Lorentzian, solid: L = 21.69 ± 0.13 nm, pass; reduced chi-squared 0.99

The certificate gives 22.3 ± 2.4 nm for this reflection. The Gaussian tops out below the 1,334-count peak and meets the background too soon, so its breadth is too small and its size too large.

In the residuals the Gaussian swings in bands of about 4σ at the top, the shoulders and the flanks; the Lorentzian’s stay near 0.

The notebook stops at the Lorentzian. The mixture of the two is the cell you write.

ASSIGNMENTS

One reflection of one pattern each, by student number

#	Pattern	hkl
1	5858	012
2	5860	020
3	5866	024
4	5860	002
5	5866	011
6	5859	013
7	5866	212

#	Pattern	hkl
8	5860	004
9	5859	030
10	5858	112
11	5866	114
12	5858	030
13	5858	213
14	5866	210

#	Pattern	hkl
15	5858	022
16	5860	211
17	5860	021
18	5858	013
19	5857	021
20	5857	110

#	Pattern	hkl
21	5860	014
22	5858	014
23	5857	020
24	5857	210
25	5857	114
26	5858	011

Your pair is beside your student number from session 4 in data/reflections.csv. In Parameters, set **PATTERN** and **REFLECTION**, such as PATTERN = "5858" and REFLECTION = "014", with the quotes, and run the Parameters cell. Section 1 is the worked example on Dataset A; section 2 is your reflection; section 3 fits the Gaussian beside your pseudo-Voigt.

AI PRACTICE: SPECIFIC PROMPTS, FROM SESSION 5

Two prompts for the same cell, typed live

A weak prompt

```
write the code for TASK 2: fit a pseudo-Voigt
to my peak and give me the crystallite size
```

In the course’s trials: the right cell, unrun, and no size. The notebook already shows the method, so the code comes out right either way.

The course’s prompt

It names the notebook and the task, the function’s parts and the order of its parameters, the names the cell must set, the weights and the bounds, and asks for the breadth and the size with a way to check them.

In the trials: an explanation of what the weights and the bounds change, the fitted numbers, and a check to run by hand.

The check is the given cell after TASK 2: it recomputes the breadth and the size from your fitted parameters. The three lines of your prompt record count toward the execution point.

AI PRACTICE: A CAUTION IS A CLAIM

On a faulted reflection, every trial answer added one caution. Is it true?

What the assistant said

A reflection the certificate marks as affected by stacking faults cannot validate the method; agreement there “may partly be luck”.

What certificate.csv says

A certified value for each such reflection, smaller than the clean ones, with the fault’s broadening already inside it. A faulted reflection is compared with its own value.

Check a caution against a document you hold before it goes into your judgment cell. The third line of your record is what you verified or rejected, and from what.

THE JUDGMENT CELL, SECTION 4

Your validation call, and one entry

(a) Your answer

- Does your method pass validation on your reflection?
- Your L ± 2 se against the certified interval with the margin, each profile’s reduced chi-squared, and what the Gaussian would have certified
- What the laboratory does next, and what a wrong call costs either way

(b) One judgment entry

- Your prompt record is the source.
- What you did: changed, rejected, or verified
- The evidence: a value from your output, or a row of the certificate table

What would you do with a pass whose margin is a few tenths of a nanometre and whose residuals run up at one edge of the window?

TO SUBMIT

Restart, Run All, save, upload by 1:30 pm on Tuesday, November 17

Scored 0 to 3

Execution scores 1 when every cell runs after a restart and the notebook holds your prompt record. The judgment cell scores 0 to 2 by the rubric in the syllabus.

Submissions are individual

Work beside a neighbour for the pair Explain cell; each of you submits your own activity.ipynb to the Session 13 assignment on Canvas.

The third hour is an office hour. Lucas grades session 13 and returns it by Tuesday, November 24.